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Detailed Course

Introduction and Motivation

General Context

The **Expectation-Maximization (EM) algorithm** is a method for **estimating latent factors** in an **unsupervised** context of conditional modeling.

Course Objectives:

- Understand **unsupervised conditional modeling** as latent factor estimation
- Develop the example of **density estimation using Gaussian Mixture Models (GMM)**
- Practice **Maximum Likelihood Estimation (MLE)**
- Understand the **alternative principle** of Expectation-Maximization for latent factor discovery

Recap: Models Used So Far

Component Models (PCA):

- Criterion based on the **variance** of centered data
- Implicit distribution: **normal**

Discriminant Models (LDA):

- Variance of projected data for within-class and between-class models

Common Point: The distribution is **fixed** (essentially normal) and we seek its parameters θ (e.g., $\theta = [\mu, \Sigma]$)

New Approach: Density Modeling

Alternative: Seek a model for the **data density**

Let $f_\theta(x) : \Omega \to \mathbb{R}$ be the **data density**

Density Estimation Methods

Overview of Methods

Non-parametric Methods:

- **k-Nearest Neighbors (kNN):** Estimation based on the k nearest neighbors
- **Parzen Windows:** Kernel-based estimation
- **RBF Networks:** Radial basis functions
- **Histograms:** Space discretization

Parametric Methods:

- **Mixture Models:** The focus of this course

Mixture Models

Definition

General Formulation

Definition:

The density $f(x)$ is generated by c "basis" functions ϕ (components), from a family $\mathcal{F}$:

$$f(x) = \sum_{j=1}^{c} \pi_j \phi(x, \theta_j)$$

Model Components:

- $\pi_j \in \mathbb{R}$: **mixture parameters**
- $\phi(x, \theta_j) \in \mathcal{F}$: **basis functions** controlled by parameters θ_j

Model Assumptions

Three Fundamental Assumptions:

1. The **number of components** $c \in \mathbb{N}^*$ is **known**
2. The **function family** $\mathcal{F}$ is **known**
3. The **labels** (class labels) are **unknown** (unsupervised context)

Probabilistic Interpretation

Probabilistic Reading of the Mixture

The density $f(x)$ represents a **random process** where x is drawn from a set of states ω_j with prior probability $\mathbb{P}(\omega_j)$.

Probabilistic Formulation:

$$f(x) = \sum_{j=1}^{c} \pi_j \phi(x, \theta_j) = \mathbb{P}(x|\theta) = \sum_{j=1}^{c} \mathbb{P}(\omega_j)\mathbb{P}(x|\omega_j, \theta_j)$$

By Identification:

- $\pi_j = \mathbb{P}(\omega_j)$ with the constraint $\sum_j \pi_j = 1$
- $\phi(x, \theta_j) = \mathbb{P}(x|\omega_j, \theta_j)$ is the **conditional probability** that x is generated by state ω_j

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Maximum Likelihood Estimation (MLE)

Likelihood Recap

Data:

Let $X = \{x_1, \dots, x_N\}$ be **unlabeled** samples generated by the mixture $f(x) = \mathbb{P}(x|\theta)$

Parameters to Infer:

$$\theta = [\pi_j, \theta_j]_{j \in [[c]]}$$

Likelihood Function

Likelihood:

Assuming the data are **independent and identically distributed (i.i.d.)**:

$$\mathbb{P}(X|\theta) \overset{\text{i.i.d}}{=} \prod_{i=1}^{N} \mathbb{P}(x_i|\theta)$$

Likelihood Estimator:

$$\hat{\theta} = \arg\max_{\theta} \mathbb{P}(X|\theta)$$

Log-Likelihood

Logarithmic Transformation:

$$LL(\theta, X) = \sum_{i=1}^{N} \log \mathbb{P}(x_i|\theta) = \sum_{i=1}^{N} \log \left[\sum_{j=1}^{c} \pi_j \phi(x_i, \theta_j) \right]$$

Maximum Log-Likelihood Estimator (MLE):

$$\hat{\theta} = \arg\max_{\theta} LL(\theta, X)$$

> **Advantage:** Transforms the product into a sum, numerically more stable

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Gaussian Mixture Model (GMM)

Choice of Gaussian Family

Since ϕ must be a **probability density function**, choosing the **normal density family** as a basis seems reasonable (an **agnostic** choice):

$$\phi \in \mathcal{F} = \{\mathcal{N}(\mu, \Sigma)\}$$

Gaussian Mixture Model:

$$f(x) = \sum_{j=1}^{c} \pi_j \mathcal{N}(x|\mu_j, \Sigma_j)$$

where $\phi_j(x) := \phi(x, \theta_j) = \mathcal{N}(x|\mu_j, \Sigma_j)$

Advantages of GMM

Universality: Can approximate **any density** (universal approximation theorem)

Linearity: Allows a **linear system** of parameters when maximizing the log-likelihood

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Simple Case: 1 Component

Direct Estimation

Parameters: $\theta = [\mu, \Sigma]$

Optimization Problem:

$$\hat{\theta} = \arg\max_{\theta} \sum_i \log e^{-(x_i-\mu)^T \Sigma^{-1} (x_i-\mu)} \Leftrightarrow \hat{\theta} = \arg\min_{\theta} \sum_i (x_i - \mu)^T \Sigma^{-1} (x_i - \mu)$$

Analytical Solutions:

$$\hat{\mu} = \frac{1}{N} \sum_i x_i$$

$$\hat{\Sigma} = \frac{1}{N} \sum_i (x_i - \hat{\mu})(x_i - \hat{\mu})^T$$

Note: Solutions identical to standard empirical estimation of mean and covariance

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Complex Case: 2 Components

Problem Statement

Parameters:

$$\theta = [\pi_1, \theta_1, \pi_2, \theta_2] = [\pi_1, [\mu_1, \Sigma_1], \pi_2, [\mu_2, \Sigma_2]]$$

with the constraint $\pi_1 + \pi_2 = 1$

Log-Likelihood:

$$LL(\theta, X) = \sum_{i=1}^{N} \log[\pi_1 \phi(x_i, \theta_1) + \pi_2 \phi(x_i, \theta_2)]$$

Difficulty: Maximization is challenging due to the **sum inside the logarithm**!

Solution: EM Algorithm

Solution: Iterative algorithm in **2 steps** (E-M) to maximize LL

$\rightarrow$ **Expectation-Maximization (EM) algorithm**

Latent Variables and EM Strategy

Introduction of Latent Variables

Fundamental Problem:

What is missing here is the **assignment** of x_i to one of the 2 components ϕ_j

Key Insight:

If we **knew** this assignment, we would treat the problem as **two instances of the 1-component case**

Binary Assignment Variables

We introduce **latent (unobserved) variables**: the binary assignment $\delta_{ij} \in \{\text{true}, \text{false}\}$ of each x_i to component ϕ_j:

- $x_i \sim \phi_1 \Rightarrow \delta_{i1} = \text{true}, \delta_{i2} = \text{false}$
- $x_i \sim \phi_2 \Rightarrow \delta_{i1} = \text{false}, \delta_{i2} = \text{true}$

But we can only **estimate** $\delta_{ij} \rightarrow$ **EM strategy**

E-step (Expectation)

E-step Principle

Parameters:

$$\theta = [\pi_1, \theta_1, \pi_2, \theta_2] = [\pi_1, [\mu_1, \Sigma_1], \pi_2, [\mu_2, \Sigma_2]]$$

Initialization:

We assume an initial value is known:

$$\theta^0 = [\pi_1^0, \theta_1^0, \pi_2^0, \theta_2^0]$$

Responsibility Calculation

We can infer the **statistical contribution** γ_{ij} of each data point x_i to each component ϕ_j (parameterized by θ_j^0):

$$\gamma_{ij}(\theta^0) = \mathbb{P}[\delta_{ij} = \text{true} | \theta^0, X]$$

Bayes' Formula:

$$\gamma_{i1}(\theta^0) = \frac{\pi_1^0 \phi(x_i, \theta_1^0)}{\pi_1^0 \phi(x_i, \theta_1^0) + \pi_2^0 \phi(x_i, \theta_2^0)}$$

Responsibility Interpretation

Formal Definition:

γ_{ij} is the **responsibility** of component j for data point i

Meaning:

- It is the **likelihood** that x_i is (purely) modeled by component ϕ_j
- γ_{ij} represents the **portion of** x_i that is modeled (generated) by component ϕ_j

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Hard vs. Soft Assignment

Hard Assignment

We can use γ_{ij} to determine $\delta_{ij} \to x_i$ can be assigned either to ϕ_1 or ϕ_2 (**maximum vote** $\to$ binarization)

$\to$ **K-means** type assignment: each data point is assigned to **one and only one** component (cluster)

Soft Assignment

EM is "softer":

A data point **contributes** (via γ_{ij}) to **multiple components** (density modes)

EM calculates a **soft assignment** with:

$$\sum_j \gamma_{ij} = 1 \quad \forall i$$

Advantage: More robust and better reflects uncertainty in assignment

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M-step (Maximization)

M-step Principle

Note: For 2 components, $\gamma_{i1} + \gamma_{i2} = 1$

Given the **responsibility** of each data point (γ_{ij}), we can estimate the parameters of each component by **weighted maximum likelihood**:

Parameter Estimation

Means:

$$\hat{\mu}_1 = \sum_{i=1}^{N} \frac{\gamma_{i1}}{\sum_{k=1}^{N} \gamma_{k1}} x_i$$

$$\hat{\mu}_2 = \sum_{i=1}^{N} \frac{\gamma_{i2}}{\sum_{k=1}^{N} \gamma_{k2}} x_i$$

Covariances:

$$\hat{\Sigma}_1 = \sum_{i=1}^{N} \frac{\gamma_{i1}}{\sum_{k=1}^{N} \gamma_{k1}} (x_i - \hat{\mu}_1)(x_i - \hat{\mu}_1)^T$$

$$\hat{\Sigma}_2 = \sum_{i=1}^{N} \frac{\gamma_{i2}}{\sum_{k=1}^{N} \gamma_{k2}} (x_i - \hat{\mu}_2)(x_i - \hat{\mu}_2)^T$$

Mixture Weights:

$$\pi_j = \frac{1}{N} \sum_{i=1}^{N} \gamma_{ij}$$

This formula ensures that $\sum_j \pi_j = 1$

EM Algorithm Convergence

Iterative Principle

The alternating **Expectation-Maximization** cycle increases the data likelihood with respect to the mixture model.

Convergence Criterion:

The process is iterated until **convergence**, i.e., when the data likelihood no longer changes (almost):

$$|LL(\theta^{t+1}, X) - LL(\theta^t, X)| < \epsilon$$

where ϵ is a tolerance threshold (e.g., $\epsilon = 10^{-6}$)

Monotonicity Property

Guaranteed Property: At each iteration, $LL(\theta^{t+1}, X) \geq LL(\theta^t, X)$

The log-likelihood **never decreases** (hill-climbing property)

EM Algorithm Limitations

Sensitivity to Initialization

Hill-Climbing: The algorithm depends on **initial parameters**

Consequences:

- **Sensitive to local maxima:** may converge to a local rather than global optimum
- Different initializations may yield different results

Initialization Strategies

Recommended Methods:

- **K-means:** use K-means clustering to initialize centers μ_j
- **K-means++:** improved K-means with smart initialization
- **Multiple Runs:** run the algorithm several times with different initializations and keep the best

Slow Convergence

Potentially slow convergence, depending on:

- Separation between components
- Distribution shapes
- Number of components

Generalization to c Components

General Latent Variables

Generalization:

$$\delta_{i1}, \delta_{i2}, \dots, \delta_{ij}, \dots, \delta_{ic}$$

where δ_{ij} = true if data point x_i is generated by component ϕ_j (δ_{ij} = false otherwise)

Responsibility:

γ_{ij}: expectation of δ_{ij} over all components

Parameters to Estimate:

$$\theta = [\pi_j, \theta_j]_{j \in [[c]]} = [\pi_j, [\mu_j, \Sigma_j]]_j$$

Complete EM Algorithm for c Components

Step 1: Initialization

Parameter Initialization:

$$\theta^0 = \{\pi_j^0, \mu_j^0, \Sigma_j^0\}_{j \in [[c]]}$$

Common Strategies:

- **General:**
  - $\pi_j = 1/c$
  - μ_j chosen randomly
  - $\Sigma_j = I_D$
- **Alternative:** use K-means for initialization

Step 2: E-step

Responsibility Calculation:

For each data point $i \in [[N]]$ and each component $j \in [[c]]$:

$$\gamma_{ij} = \frac{\pi_j \phi(x_i, \theta_j)}{\sum_{k=1}^{c} \pi_k \phi(x_i, \theta_k)}$$

Step 3: M-step

Mixture Parameter Estimation:

Means:

$$\mu_j = \frac{\sum_{i=1}^{N} \gamma_{ij} x_i}{\sum_i \gamma_{ij}}$$

Covariances:

$$\Sigma_j = \frac{X_j \Gamma_j X_j^T}{\mathrm{Tr}(\Gamma_j)}$$

where $\Gamma_j = \mathrm{diag}(\gamma_{1j}, \ldots, \gamma_{Nj})$ and X_j is centered on μ_j

Weights:

$$\pi_j = \frac{\sum_i \gamma_{ij}}{N}$$

Step 4: Iteration

Iterate steps 2 and 3 until convergence

Practical Modeling

Parameters Influencing Convergence

The **a priori parameterization** of the mixture affects convergence:

Critical Parameters:

1. c**:** number of components
2. Σ_j**:** form of covariance matrices (diagonal, full, parameterized)

Risks:

- **Too flexible** or **too rigid** models $\rightarrow$ poor convergence or lack of convergence
- **Over-parameterization:** $D \times D \times c + 2 \times c$ variables

  If N is small, D large, and c large $\rightarrow$ **problematic over-parameterization**

Covariance Matrix Form

Over-Parameterization Problem

Dimension of Σ:

$\Sigma \in \mathbb{R}^{D \times D}$

Critical Example:

Character recognition with $x_i \in \mathbb{R}^{256}$

$\rightarrow$ Need to estimate $256^2 \times c$ parameters for covariance (with about 7000 data points)!

Parameterization Strategies

Spherical Models:

- $\Sigma = \sigma I_D$
- **1 parameter** per component

Diagonal Models:

- $\Sigma = \text{diag}(\sigma_1, \ldots, \sigma_D)$
- D **parameters** per component

Full Models:

- $\Sigma \in \mathbb{R}^{D \times D}$
- $O(D^2)$ **parameters** per component

Practical Advice: Start with simple models (spherical) then increase complexity if needed

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Preprocessing: Dimensionality Reduction with PCA

High-Dimensional Problem

For high-dimensional data (e.g., $D = 256$), complex models do not converge because the number of parameters p is too large.

Solution: Prior PCA

Recommended Strategy:

1. Apply **PCA** to reduce dimension from D to K dimensions
2. Apply **EM** in the space of the first K principal components

Practical Example:

- 50 principal components reconstruct 90% of the signal
- EM in the space of 50, 10, or 2 principal components

Advantages:

- Drastic reduction in the number of parameters
- Noise elimination
- Faster and more stable convergence

Selecting the Number of Components

Parsimony Principle

Problem:

The larger c, the fewer points can be assigned to each component (on average)

Objective:

Seek **parsimonious models**, i.e., with a **small number of parameters** to estimate

Bayesian Information Criterion (BIC)

Likelihood-Complexity Trade-off:

- The larger c, the better the likelihood estimation LL
- But the more complex the model (many parameters)

BIC Formula:

$$BIC(\theta, X) = -2 \cdot LL(\theta, X) + |\theta| \log(N)$$

where $|\theta|$ is the **number of parameters** in the model

Selection Criterion:

$$\hat{\theta} = \arg\min_{\theta} BIC(\theta, X)$$

BIC Components:

- $-2 \cdot LL(\theta, X)$: penalizes models with poor likelihood
- $|\theta| \log(N)$: penalizes model complexity

Interpretation: BIC favors models that explain data well **without overfitting**

Model Selection Limitations

Practical Challenges:

- Need to **test all models** (different values of c)
- Depends on **convergence** of each model
- Often requires **fine manual tuning**

Synthesis of Key Concepts

Gaussian Mixture Models (GMM)

Generalization:

The Gaussian mixture model **generalizes** the assumptions underlying PCA and LDA

Explicit Modeling and Estimation:

- **Explicit modeling** of density
- **Estimation** by maximum likelihood

Unsupervised Classification:

If components are classes, data point x_i is associated with class j for which:

$$\mathbb{P}(C = j | x_i) \approx \pi_j \phi_j(x_i)$$

is maximized among all classes

EM Algorithm

Characteristics:

- **Iterative algorithm** to maximize (log-)likelihood
- Principle used in **many other scenarios**
- Based on defining **unobserved (latent) variables** δ_{ij}

Applications:

- **Probabilistic Latent Semantic Analysis (pLSA):** EM where hidden variables are latent concepts
- **Hidden Markov Models (HMM)**
- **Matrix Factorization**

Relationship Between GMM and K-means

Conceptual Comparison

K-means:

- **Hard assignment**
- Each point belongs to **one cluster only**
- Minimizes Euclidean distance to centers

GMM with EM:

- **Soft assignment**
- Each point has a **probability of belonging** to each component
- Maximizes likelihood according to a probabilistic model

GMM as a Generalization of K-means

K-means is a special case of GMM when:

- Covariance matrices $\Sigma_j = \sigma^2 I$ (spherical and identical)
- $\sigma \to 0$: responsibilities γ_{ij} become binary
- Soft assignment $\to$ hard assignment

Typical Exam Questions

Conceptual Questions

Gaussian Mixture Model:

- What is a Gaussian mixture model (GMM)?
- Why can it be seen as conditional modeling?
- How can it be used to generate data?

Responsibility:

- What is responsibility?
- How to interpret it?

EM Algorithm:

- Why is EM for GMM a MLE?
- How to interpret the E-step?
- How to interpret the M-step?

Technical Questions

Relationship with K-means:

- Discuss the relationship between GMM and K-means
- What are hard and soft assignments?

Convergence and Optimization:

- How to choose the number of components?
- What are the initialization strategies?
- How to handle over-parameterization?

Mathematical Developments

Important Advice: It is **strongly recommended** to develop the algebra contained in this chapter

Key Points to Master:

- Derivation of the log-likelihood
- Calculation of responsibilities using Bayes' rule
- Derivation of M-step update equations
- Proof of LL monotonicity in the EM algorithm

Mathematical Concepts to Master

Probability and Statistics

Fundamentals:

- Multivariate normal distribution
- Maximum likelihood estimation (MLE)
- Log-likelihood
- Bayes' rule

Modeling:

- Latent random variables
- Conditional probabilities
- Conditional expectation

Optimization

Techniques:

- Iterative optimization
- Hill-climbing algorithms
- Convergence criteria
- Local vs. global maxima

Selection Criteria:

- BIC (Bayesian Information Criterion)
- Bias-variance trade-off
- Model parsimony